

Title: DALScooter

Team Size: ~4 Members

Project Length: Approx. 2.5 months

Project Leader: Sigma Jahan



Objective:

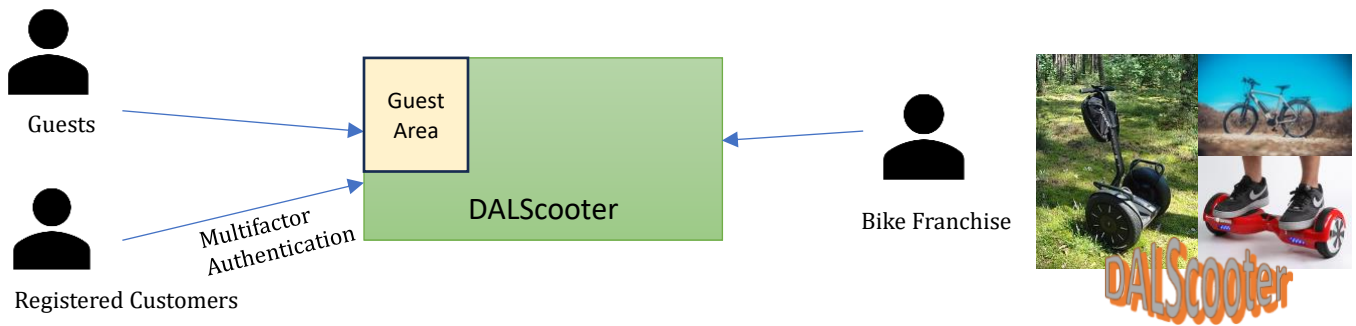
The primary objective of this project is to build cloud plumbing system, where an application will be designed using serverless technologies to process data (more specifically on-demand data). It should not require any specialist for using the implemented system. In this project, you will be building a “cloud data plumbing system”, which could be used by many clients to process their data. You will use different backend services, and simple front-end application to build the application. This project is introduced in the Serverless Data Processing Course (CSCI 5410) to fulfill the course requirement. This is a group project (weightage 40%), and each group is required to perform specific tasks within a given time frame. There are project constraints, and scope, which must be followed by each team. The project has multiple deliverables, which must be completed at the end of each sprint.

Plagiarism Policy:

- This project is a group task. Collaboration of any type (outside the assigned group) amounts to a violation of the academic integrity policy and will be reported to the AIO.
- Content cannot be copied verbatim from any source(s). Please understand the concept and write in your own words. In addition, cite the actual source. Failing to do so will be considered as plagiarism and/or cheating.
- AI (Artificial Intelligence) tools cannot be used for any part of the project component. AI policy mentioned in syllabus must be followed.
- The Dalhousie Academic Integrity policy applies to all material submitted as part of this course. Please understand the policy, which is available at:
https://www.dal.ca/dept/university_secretariat/academic-integrity.html

Hypothetical Scenario:

DALSoft5410 is building a serverless DALScooter application using multi-cloud deployment model, and backend-as-a-service (BaaS). The “DALScooter”, should provide customization feature, and additional services for authorized users, and limited services to guests. The DALScooter should provide an online virtual assistance, which can quickly answer the queries of users, and in addition, the application should provide a message passing functionality between the authorized users and agents.



Use Case:

There will be three different types of users in the DALScooter system.

(1) **Guests**, (2) **Registered Customers**, and (3) **Bike Franchise** (They are super users or admin users).

Guest Users

- The guest users should be able to visit the application site, and check availability and tariffs of different types of bikes
- type1: Gyroscooter or type2: eBikes or type3: Segway
- The guest users will be able to use virtual assistant or the chatbot to get basic navigation information.
- The guest users should be able to view the feedback and polarity of the overall feedback received for a particular bike or scooter (gyroscooter or eBike or segway)

Registered Customers

- Registered Customers will be able to perform all the operations that a guest user can perform. In addition, the following points will be applicable:
- A registered customer will get a notification once the registration process is completed.
- The registered customers will access the system using multi-factor authentication. 1st factor is userID/password, 2nd factor is Question-Answer, and 3rd factor is solving clue using Caesar cipher. Each factor must be validated in a sequential manner.
- The registered user will also get a notification after a successful sign-in process.
- Once a registered customer enters the system, they should be able to reserve an e-bike or gyrocopter scooter for a specific period (from a start time to an end time. You can consider only daily booking).
- Once logged-in, a customer can also use the virtual assistant to
 - navigate within the site.
 - obtain the bike access code and duration of use based on booking reference code.
 - Support customer to Bike franchise communication
- Registered customer should be able to provide feedback using a standard form about their usage of eBike or gyroscooter or segway.

Franchise

- Franchise operators will be able to perform all the operations that a guest user can perform. In addition, franchise operators should be able to do the following:
- A franchise operator will get a notification once their registration process is completed.
- The franchise operator are admin users, but they will also access the system using multi-factor authentication. 1st factor is userID/password, 2nd factor is Question-Answer, and 3rd factor is solving clue using Caesar cipher. Each factor must be validated in a sequential manner.
- The franchise operator will also get a notification after a successful sign-in process.
- Once a franchise operator enters the system, they should be able to add/update an ebike or gyroscooter or segway, such as adding a new bike with an access code, adding more features {height adjustment, higher battery life etc.}, updating hourly rental rate, adding a discount code etc.
- Once logged-in, a franchise operator can also use the virtual assistant to
 - navigate within the site.
 - obtain the bike number and duration of use based on booking reference code.
- Franchise operator should be able to communicate with the customer once a issue_ticket is received. This is an asynchronous communication. It can also be designed as a synchronous communication as an additional feature.

TASK DETAILS

Module Details:

You must use Serverless architecture to build the following modules.

1. User Management & Authentication Module:

Sign up with validation and multi-factor authentication using AWS – Perform the user registration through your light-weight front end application, and use the following services:

- User details should be stored in DynamoDB.
- User ID/Password (1st factor authentication) – AWS Cognito
- Question/Answer (2nd factor authentication) – DynamoDB + AWS Lambda
- Caesar cipher (3rd factor authentication) – AWS Lambda + DynamoDB

2. Virtual Assistant Module:

Bots should respond to queries (AWS Lex + DynamoDB + AWS Lambda) **OR** (GCP DialogueFlow + Firestore + Cloud Function) **OR** (AzureBot Service CosmosDB + Azure Function)–

- Online virtual assistance for navigation within the application site. Such as, answering “how to register?”
- searching bike access code and usage duration based on booking reference code.
- Accepts customer concerns and forwards to franchise operator.

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3. **Message Passing Module:**

Message passing module (GCP pub/sub or AWS SNS/SQS) will be used for message passing between parties.

- Customers will submit the concerns based on booking reference code, which will be published using Publisher, and a function subscribed to message will forward it to a random franchise operator for answering.
- This entire communication should be logged in a NoSQL database like, DynamoDB or Firestore.

4. **Notifications:**

Notification module will use AWS SNS and AWS SQS

- The service will be used for sending successful registration, and successful login notifications to the registered user's email.
- The notification service will also be used for sending booking confirmation and booking failure. **Note:** Payment processing is not within the scope of the project. We assume payment will be done in-person after a successful booking, therefore, ignore building payment module.
- SNS – SQS should be used between application components, such as - “eBike Booking Request” and “eBike Booking Approval” process. Booking request will enter queue (Standard SQS), which will trigger Lambda for performing the approval task.

5. **Data Analysis & Visualization:**

Product of this module is used by all types of users.

- Franchise Operator can check total users, and login statistics on the admin page within frontend (LookerStudio dashboard or QuickSight or PowerBI- maybe embedded within a frame)
- Guests, Customers, and Franchise Operator should be able to see the customers' feedbacks in a tabular format within the frontend retrieved from DynamoDB OR S3 bucket OR Firestore OR GCP storage OR CosmosDB.
- The feedback will automatically be analyzed in the backend to understand the sentiment and present in the frontend for all types of users. Use any natural language API available.

6. **Web Application Building and Deployment:**

Build a front-end application using suitable framework such as, React and use it to call all the backend services –.

- Hosting of entire application and user/client facing interface should be done using GCP CloudRun or AWS Fargate or similar services available in AWS, GCP or Azure.
- Use CloudFormation or GCP Cloud Deployment Manager for automation of service deployment.

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General Tasks:

- Testing Module (GCP + AWS) – Test cases must be validated. Use a table to capture the test cases and results of the testing.
- Check for errors and border line cases.
- In addition to test cases your team decides, you also need to perform the following tests:
 - Registered users' successful authentication and guests' users failed login,
 - Data accuracy or integrity test in upload-retrieval process to/from database,
 - Chatbot response test for valid and invalid utterances
 - API testing for reliability and functionality check,
 - Lambda or Cloud Function testing using test events and inspecting logs,
- Provide evidence of testing in the form of screenshots
- To maintain code quality – focus on code readability, code reusability, code cleaning, commenting.
- Perform regular git commits. Expected mainly during Sprint 2 and Sprint 3.
- Use any standard project management tool for better management of resources, and progress tracking.

Deliverables – Project (40% of Course Grade)

Sprint 1 (25% of Project Grade): Point A, and Point B will be graded.

- A. Submit a report on project planning, document meeting agenda, background research related to cloud platforms/ services, and/or required programming.
- B. It should be a single 5-page report submitted by each Team.
- C. The report is due on **Jun 6, 2025** - with details mentioned in **point A**, and **point B**.
- D. **Additional Submission required:** All team meetings (video and text logs) to be submitted separately.
- E. Attend non-evaluative Q&A session with **Project Lead** between **May 26** and **Jun 8** and explain individual contribution and challenges.

Sprint 2 (25% of Project Grade):

- A. Submit a 4-page report (minimum) by each team with details mentioned in point B.
- B. Project planning, overall system architecture design, details of completed implementation of at least 2 modules with functional and/or validation testing of those modules (such as, "Registration", "Virtual Assistant"). Pseudo-code, flowchart/ activity diagram of completed modules. Provide Gantt chart for all other modules and next sprint.
- C. The report is due on **Jul 4, 2025** - with details mentioned in point A, and point B.

- D. Additional Submission required: All team meetings (video and text logs) to be submitted separately. Flowcharts/ Architecture/ test case table etc. must be submitted separately for clarity.
- E. Meeting with **Project Lead** is NOT Required for Sprint 2. However, your team can meet the Project Lead to clarify doubts or informal verification.

Sprint 3 (25% of Project):

- A. Submit a 4-page report by each team with details mentioned in point B.
- B. Final system architecture design, details of all pending implementations of the modules with functional and/or validation testing of those modules. Pseudo-code, flowchart/ activity diagram of all completed or partially completed modules.
- C. This is the final report, which is due on **Aug 1, 2025** - with details mentioned in point A, and point B.
- D. **Additional Submission required:** All team meetings (video and text logs) to be submitted separately. Flowcharts/ Architecture/ test case table etc. must be submitted separately for clarity.

Final Q&A session (10% of Project):

- A. 20 min to 30 min final Q&A session. **Project Lead** and/or **Instructor** will ask questions related to the project modules and concepts.
- B. Bookable Calendar will be shared for the Q&A.
- C. It should be done within this timeframe **Jul 25 to Aug 5, 2025** (20 min/group)

Final Project Presentation or DEMO (15% of Project):

- A. Each team needs to submit a 45 min (40 min+5min) recorded video with the following -
- B. **40-min:** In this total 40-min time slot, each team member needs to explain their individual contributions with audio, video, and screenshare. The recording should clearly highlight how they achieved the task, business logic, coding structure, configurations etc. These videos do not require editing or merging.
- C. **5-min:** In this video the screen needs to be captured to show different use cases of the application. Verbal explanations are not needed in this 5-min video.
- D. Demo videos are due on **Aug 1, 2025**

PROJECT REQUIREMENT DOCUMENT – CSCI 5410/S25

[Project sprint reports evaluation will be based on project rubric given below]

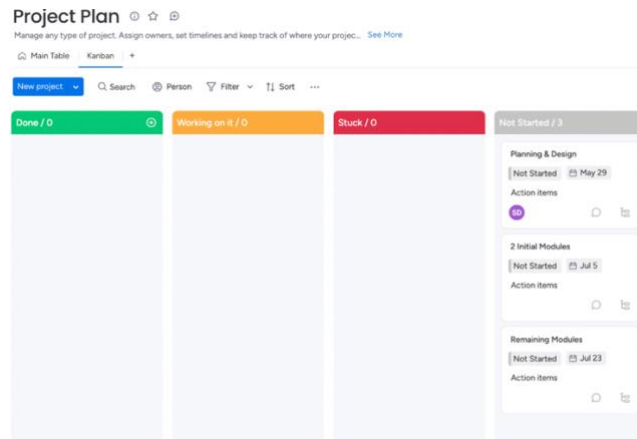
Project Report Evaluation Rubric - based on the discussion board rubric (McKinney, 2018)

	Excellent (20%)	Proficient (15%)	Marginal (10%)	Unacceptable (0%)
Completeness including Citation	All required tasks are completed	Submission highlights tasks completion. However, missed some tasks in between, which created a disconnection	Some tasks are completed, which are disjoint in nature.	Incorrect and irrelevant
Correctness	All parts of the given tasks are correct	Most of the given tasks are correct However, some portions need minor modifications	Most of the given tasks are incorrect. The submission requires major modifications.	Incorrect and unacceptable
Novelty	The submission contains novel contribution in key segments, which is a clear indication of application knowledge	The submission lacks novel contributions. There are some evidences of novelty, however, it is not significant	The submission does not contain novel contributions. However, there is an evidence of some effort	There is no novelty
Clarity	The written or graphical materials, and developed applications provide a clear picture of the concept, and highlights the clarity	The written or graphical materials, and developed applications do not show clear picture of the concept. There is room for improvement	The written or graphical materials, and developed applications fail to prove the clarity. Background knowledge is needed	Failed to prove the clarity. Need proper background knowledge to perform the tasks
Group Work	Evidence of group work, meeting logs, Coordination	Evidence of group work. However, missed meeting logs, room for improvement in Coordination	Missed meeting logs, failed to display group coordination	No group work done. Project is unacceptable

Note: Final Q&A and Presentation or DEMO will not follow the above rubric

Project Expectations:

- Please respect every individual and follow the “Culture of Respect” section provided in the course syllabus.
- All members should contribute equally to the project.
- One or two member(s) of a team completing the project cannot be considered as a successful group project, and the entire group may receive lower score.
- All members of a team may not receive similar score. Each sprint will have individual contribution grade as well.
- Try to use some form of project management software to track your project status and timeline. E.g.



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